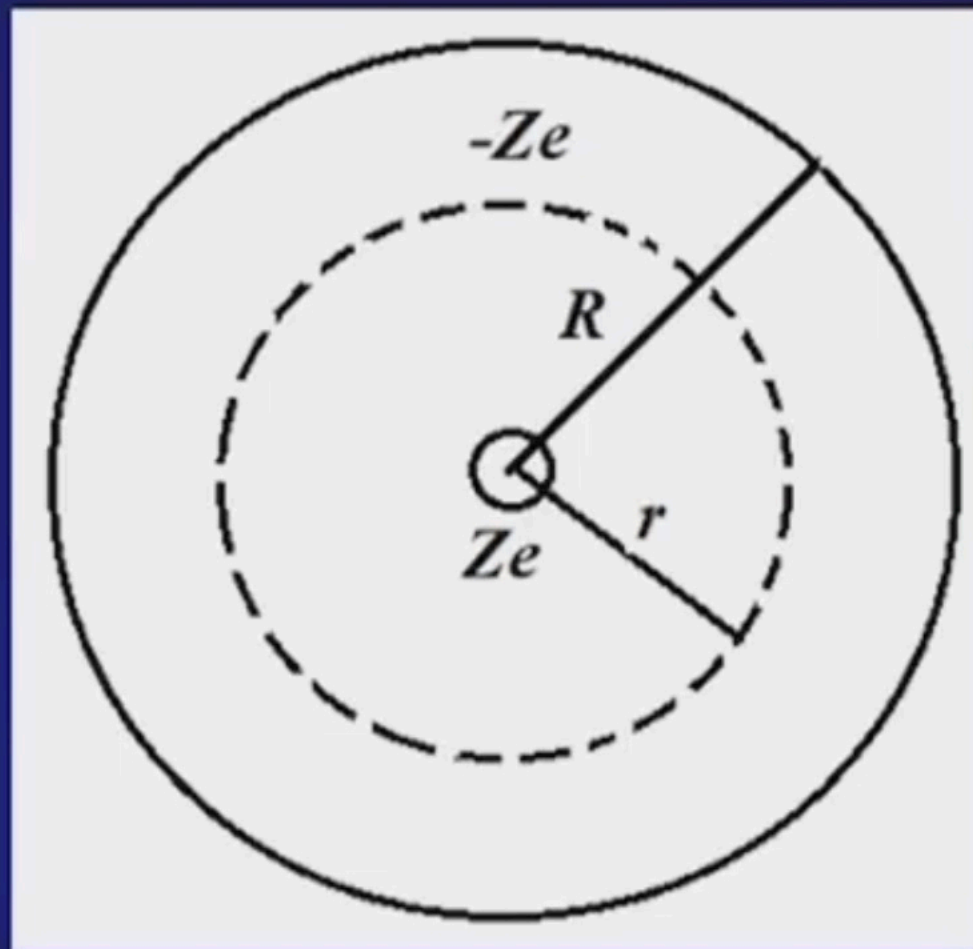


2020年期中考试卷答案

I. Fill in the space underlined (50% in total)

1. In a 1911 paper, Ernest Rutherford said: In order to form some idea of the forces required to deflect an α particle through a large angle, consider an atom containing a point positive charge Ze at its center and surrounded by a distribution of negative electricity $-Ze$ uniformly distributed within a sphere of radius R . The electric field E for a point inside an atom at a distance r from the center is _____.



$$\oiint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$$

$$q = Ze - \frac{4}{3}\pi r^3 \cdot \frac{Ze}{\frac{4}{3}\pi R^3} = Ze\left(1 - \frac{r^3}{R^3}\right)$$

$$E \cdot 4\pi r^2 = \frac{Ze}{\epsilon_0} \left(1 - \frac{r^3}{R^3}\right)$$

$$E = \frac{Ze}{4\pi\epsilon_0} \left[\frac{1}{r^2} - \frac{r}{R^3} \right]$$

2. As shown in Fig. 1, a charge per unit length λ is distributed uniformly along a thin rod of length L . The potential (chosen to be zero at infinity) at a point P a distance y from one end of the rod and in line with it is $V_P =$ _____. The component of the electric field at P in the y direction (along the rod) is $E_y =$ _____. The component of the electric field at P in a direction perpendicular to the rod is $E_x =$ _____.

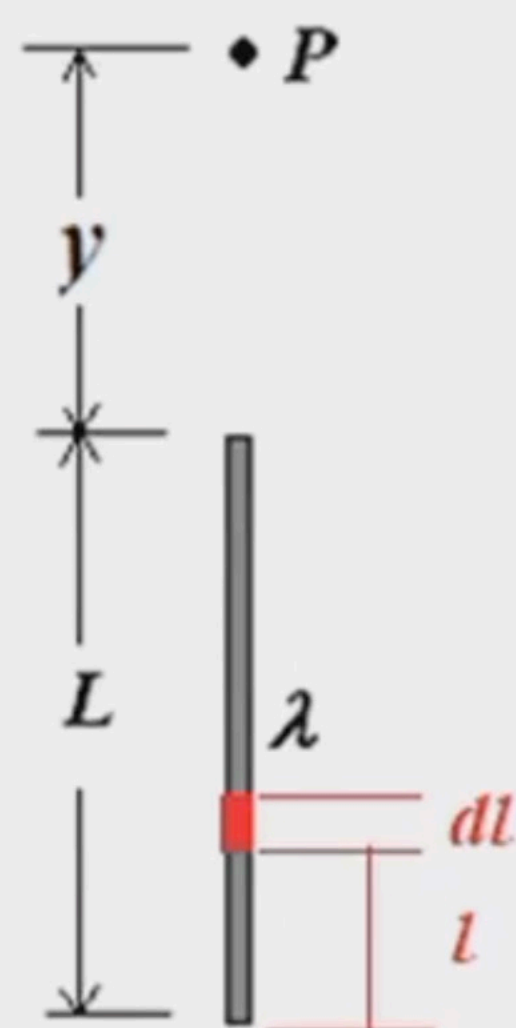


Figure 1

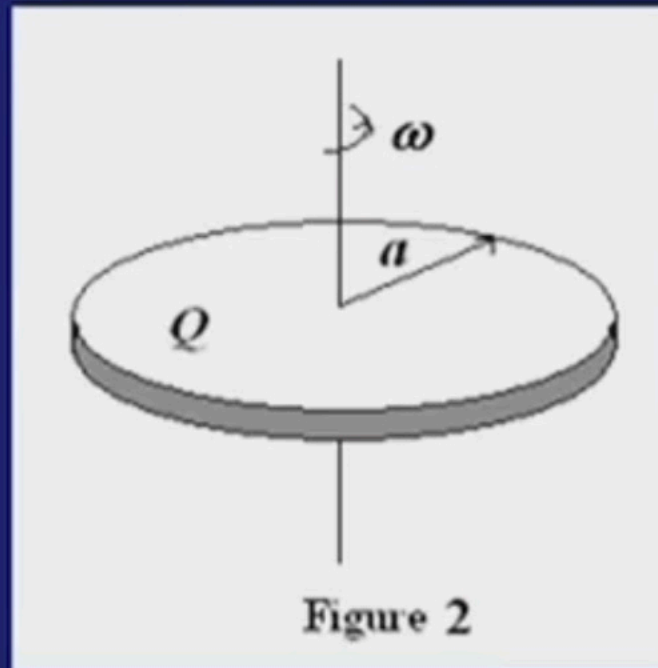
$$dV = \frac{\lambda dl}{4\pi\epsilon_0(L-l+y)}$$

$$V_P = \int_0^L \frac{\lambda dl}{4\pi\epsilon_0(L-l+y)} = -\frac{\lambda}{4\pi\epsilon_0} \ln(L-l+y) \Big|_0^L = \frac{\lambda}{4\pi\epsilon_0} \ln \frac{L+y}{L}$$

$$E_y = -\frac{\partial V_P}{\partial y} = -\frac{\lambda}{4\pi\epsilon_0} \left[\frac{1}{L+y} - \frac{1}{y} \right] = \frac{\lambda}{4\pi\epsilon_0} \frac{L}{y(L+y)}$$

$$E_x = -\frac{\partial V_P}{\partial x} = 0$$

3. There is a thin disk of radius a , with total charge Q uniformly distributed on its surface, rotating at a constant angular speed ω , as shown in Fig. 2. What is the magnetic dipole moment (磁偶极矩)?

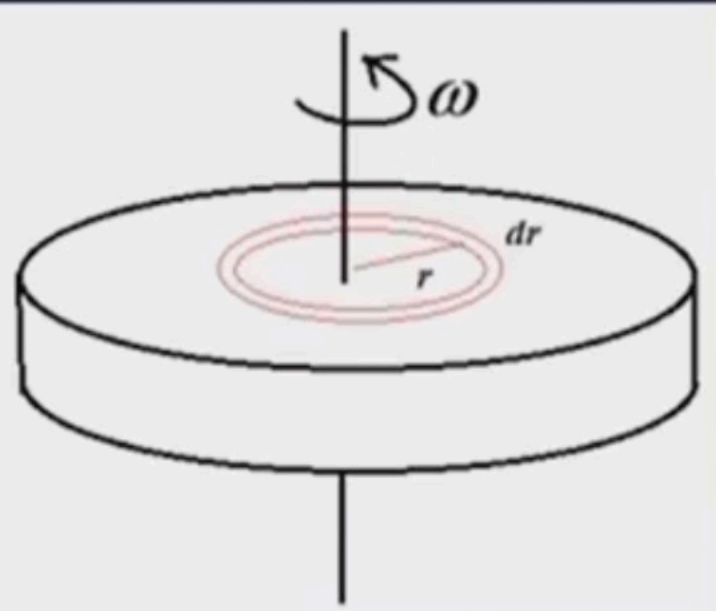


$$dq = \frac{Q}{\pi a^2} \cdot 2\pi r dr$$

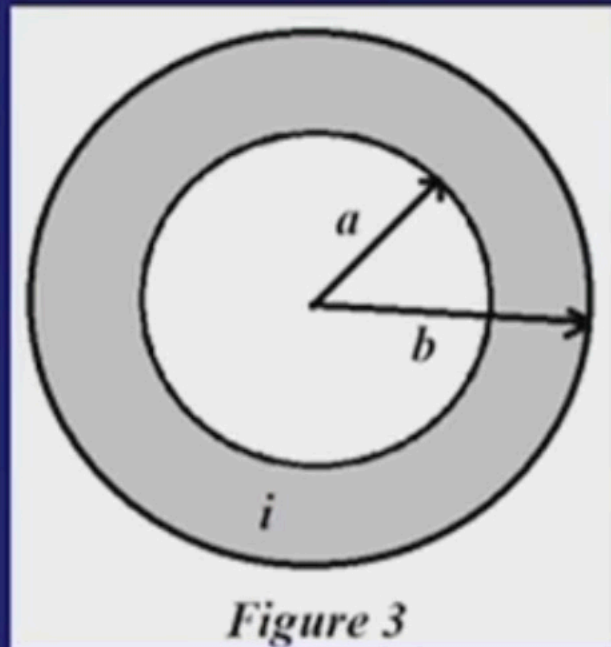
$$di = \frac{dq}{T} = \frac{\omega}{2\pi} \cdot \frac{Q}{\pi a^2} \cdot 2\pi r dr = \frac{Q\omega}{\pi a^2} r dr$$

$$d\mu = \pi r^2 di = \frac{Q\omega}{a^2} r^3 dr$$

$$\mu = \int_0^a \frac{Q\omega}{a^2} r^3 dr = \frac{1}{4} Q\omega a^2$$



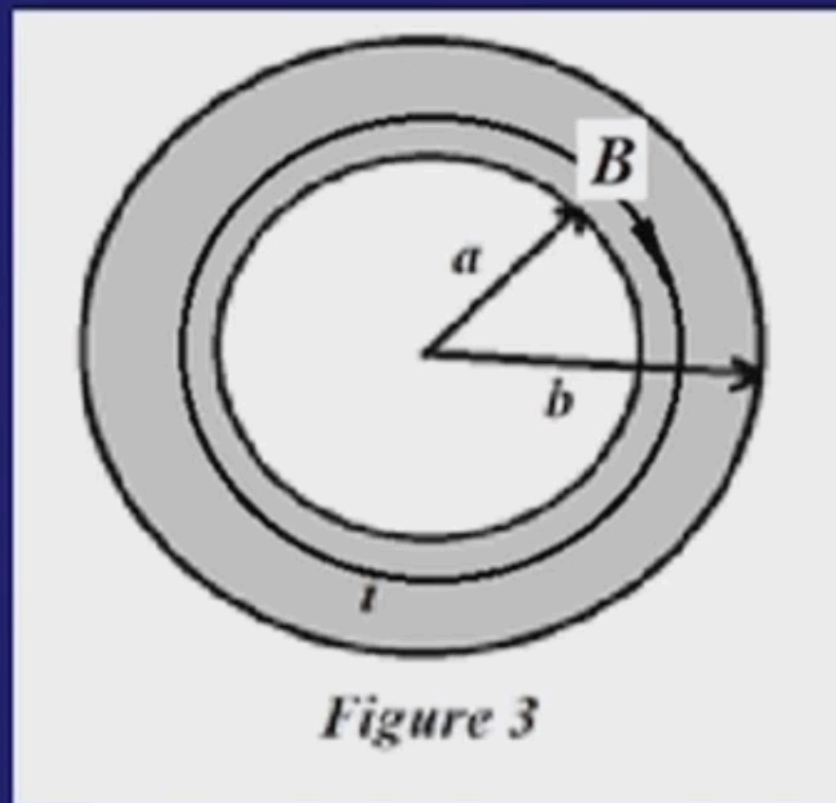
4. Figure 3 shows a cross section of a hollow, cylindrical conductor of radii a and b , carrying a uniformly distributed current i . The $B(r)$ for the range $a < r < b$ is given by _____.



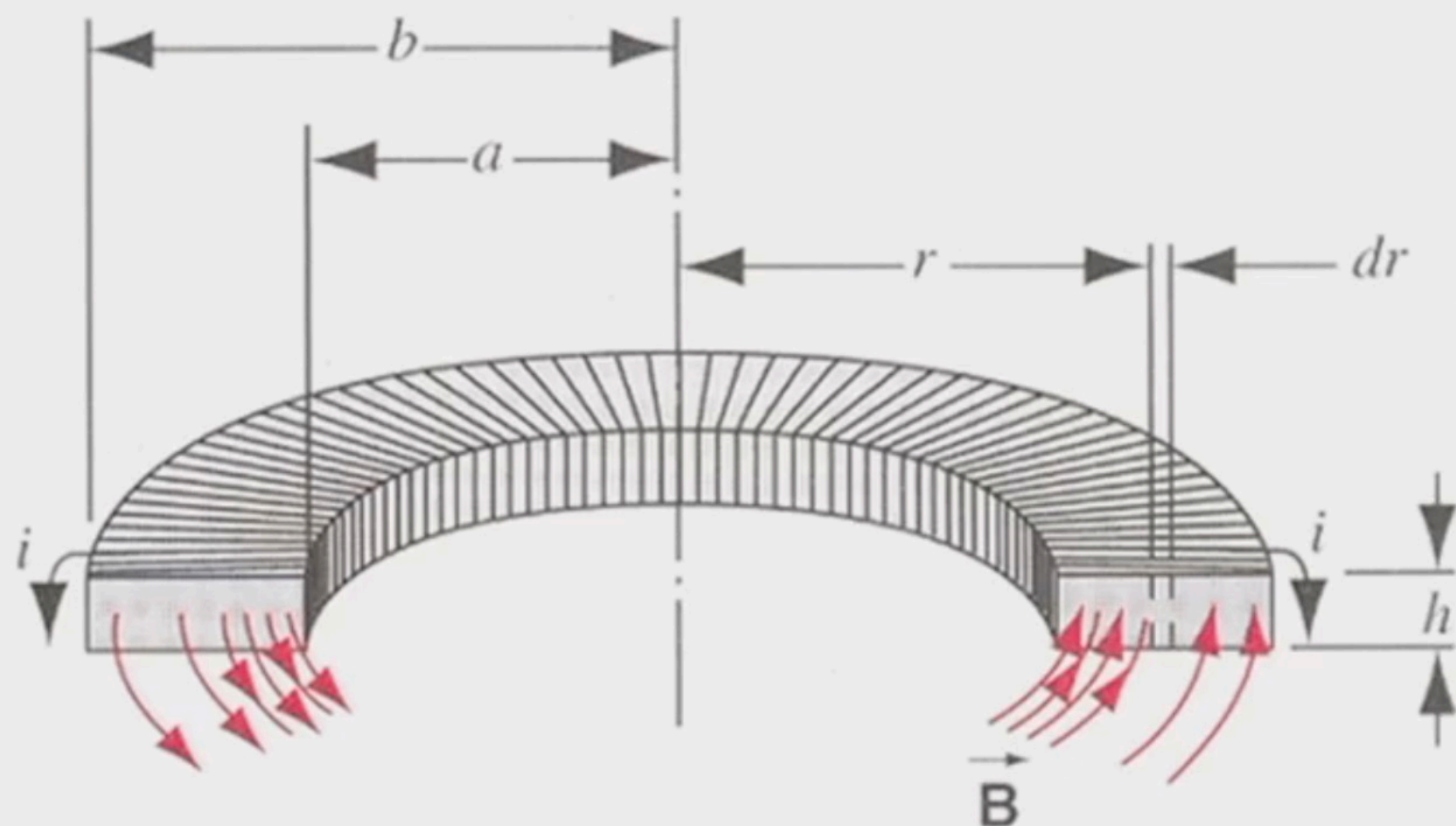
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$$B \cdot 2\pi r = \mu_0 \frac{i}{\pi(b^2 - a^2)} [\pi(r^2 - a^2)]$$

$$B = \frac{\mu_0 i}{2\pi r} \cdot \frac{r^2 - a^2}{b^2 - a^2}$$



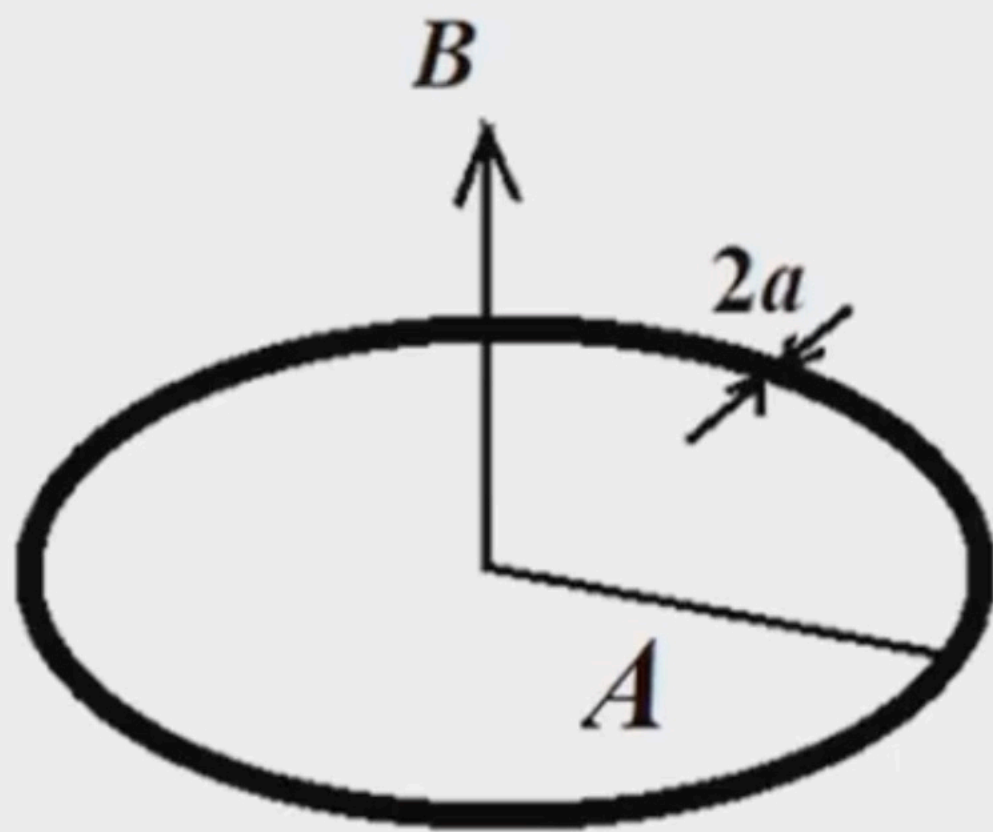
5. Figure 4 show a toroid of rectangular (矩形螺绕环), with inside radius a and outside radius b , and height h . The inductance (自感 L) is given by



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$
$$B \cdot 2\pi r = \mu_0 Ni$$
$$B = \frac{\mu_0 Ni}{2\pi r}$$

$$\psi = N \int_a^b B h dr = N \int_a^b \frac{\mu_0 N i h}{2\pi r} dr = \frac{\mu_0 N^2 i h}{2\pi} \ln \frac{b}{a}$$
$$L = \frac{\psi}{i} = \frac{\mu_0 N^2 h}{2\pi} \ln \frac{b}{a}$$

6. A uniform magnetic field B is changing in magnitude at a constant rate dB/dt . You are given a mass m of copper that is to be drawn into a wire and formed into a circular loop. Assuming B perpendicular to the loop and the copper resistivity (电阻率) is of ρ and the density (密度) of δ , then the induced current in the loop is given by _____,



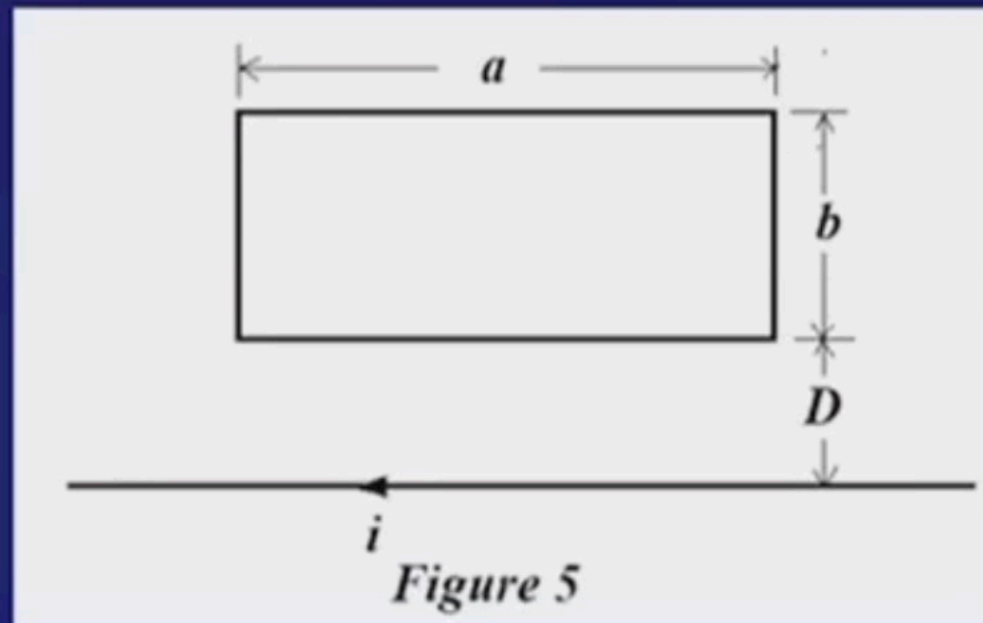
$$R = \rho \frac{2\pi A}{\pi a^2} = \rho \frac{2A}{a^2}$$

$$m = 2\pi A \cdot \pi a^2 \cdot \delta = 2\pi^2 a^2 A \delta$$

$$A = \frac{m}{2\pi^2 a^2 \delta}$$

$$i = \frac{\mathcal{E}}{R} = \frac{\pi A^2 \frac{dB}{dt}}{\rho \frac{2A}{a^2}} = \frac{\pi A a^2}{2\rho} \frac{dB}{dt} = \frac{\pi \frac{m}{2\pi^2 a^2 \delta} a^2}{2\rho} \frac{dB}{dt} = \frac{m}{4\pi\rho\delta} \frac{dB}{dt}$$

7. A rectangular loop of wire with length a , width b , and resistance R is placed near an infinitely long wire carrying current i , as shown in Fig. 5. The distance from the long wire to the loop is D . The current in the loop is given by _____, as it moves away from the long wire speed v .



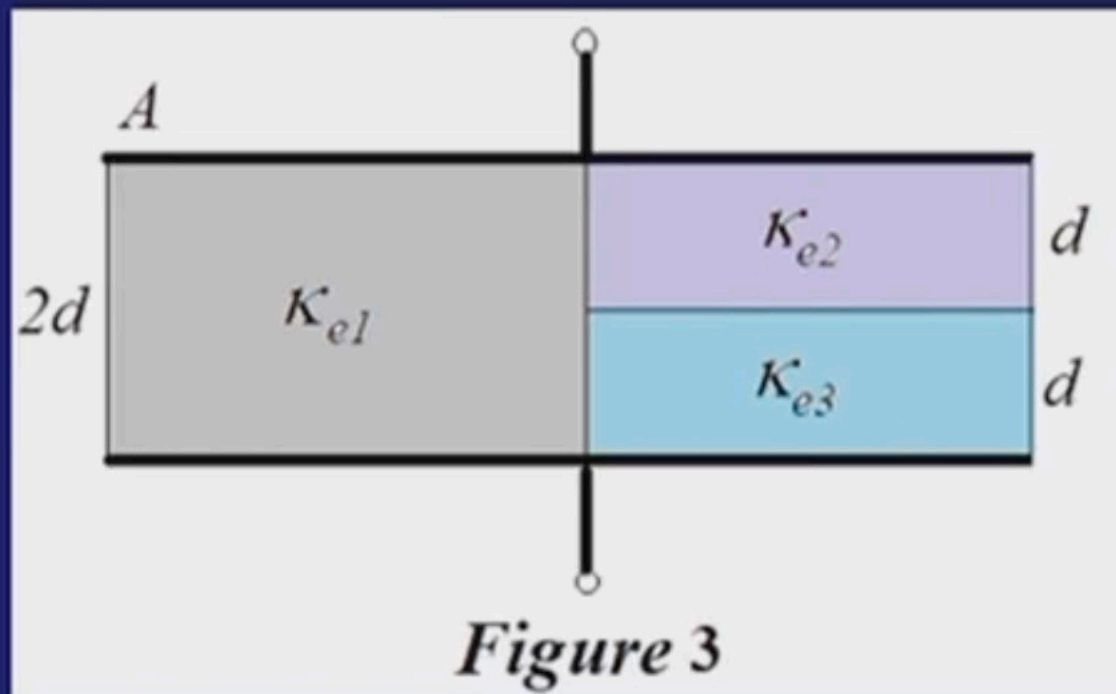
$$B = \frac{\mu_0 i}{2\pi r}$$

$$\psi = \int_D^{D+b} \frac{\mu_0 i}{2\pi r} a dr = \frac{\mu_0 i a}{2\pi} [\ln(D+b) - \ln D]$$

$$\varepsilon = -\frac{d\psi}{dt} = -\frac{\mu_0 i a}{2\pi} \left[\frac{1}{D+b} - \frac{1}{D} \right] \frac{dD}{dt} = \frac{\mu_0 i a}{2\pi} \frac{b}{D(D+b)} v$$

$$i = \frac{\varepsilon}{R} = \frac{\mu_0 i}{2\pi R} \frac{abv}{D(D+b)}$$

8. A parallel-plate capacitor with area A , distance $2d$, is filled with three dielectrics (电介质) as shown in Fig.6, the capacitance (电容 C) is given by



$$C_1 = \kappa_{e1} \frac{\varepsilon_0 A / 2}{2d} = \frac{\kappa_{e1} \varepsilon_0 A}{4d},$$

$$C_2 = \kappa_{e2} \frac{\varepsilon_0 A / 2}{d} = \frac{\kappa_{e2} \varepsilon_0 A}{2d}$$

$$C_3 = \kappa_{e3} \frac{\varepsilon_0 A / 2}{d} = \frac{\kappa_{e3} \varepsilon_0 A}{2d}$$

$$C = C_1 + \frac{C_2 C_3}{C_2 + C_3} = \frac{\varepsilon_0 A}{4d} \left[\kappa_{e1} + \frac{2\kappa_{e2} \kappa_{e3}}{\kappa_{e2} + \kappa_{e3}} \right]$$

II. Problems (Present the necessary equations in solution) (50%)

1. (12%) As shown in Figure 7, a wireframe (线框) with current i_2 is composed of an arc (圆弧) with the flare angle (张角) of $2(\pi - \varphi)$ and the radius of a , and a subtense (弦) connecting two ends of the arc. At the same time, there is a long straight conducting wire with current i_1 , which is located at the center of the arc and perpendicular to the plane of the wireframe. Please calculate the torque (力矩) τ acted on the wireframe.

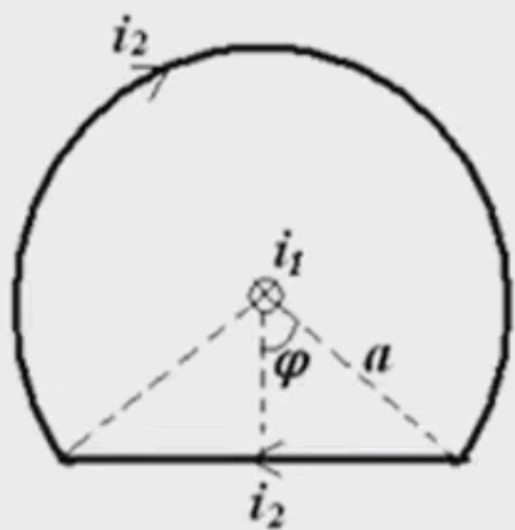
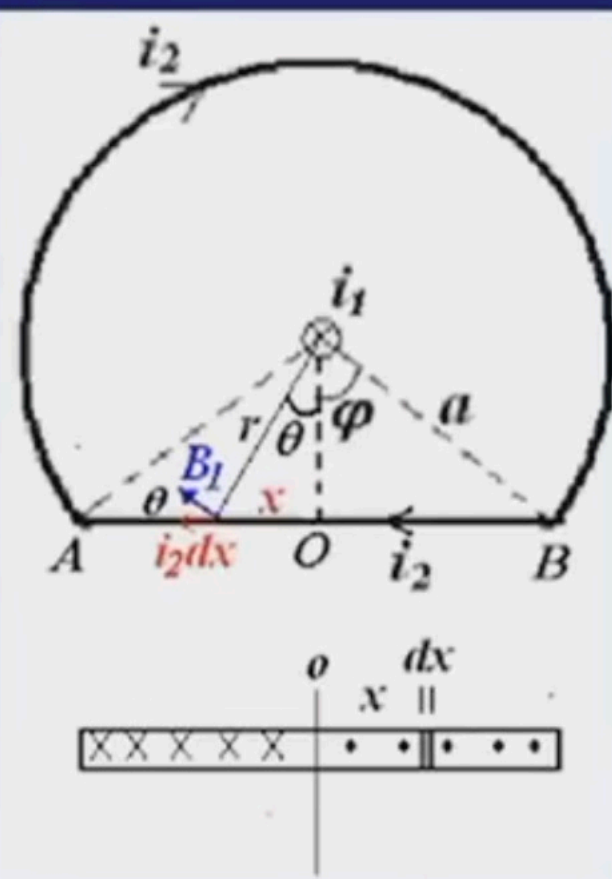


Figure 7



The magnetic field induced by i_1 : $B = \frac{\mu_0 i_1}{2\pi r}$

The torque on the AB arc: $\tau = 0$

The torque on the AB subtense: $\tau \neq 0$

$$d\vec{F} = i d\vec{s} \times \vec{B}, \quad dF = i_2 dx \cdot B \sin \theta = \frac{\mu_0 i_1 i_2 dx}{2\pi r} \sin \theta$$

$$d\tau = x \cdot dF = \frac{\mu_0 i_1 i_2 \sin \theta}{2\pi r} x dx$$

$$x = a \cos \varphi \cdot \tan \theta, \quad dx = a \cos \varphi \cdot \frac{d\theta}{\cos^2 \theta}$$

$$r = \frac{a \cos \varphi}{\cos \theta}$$

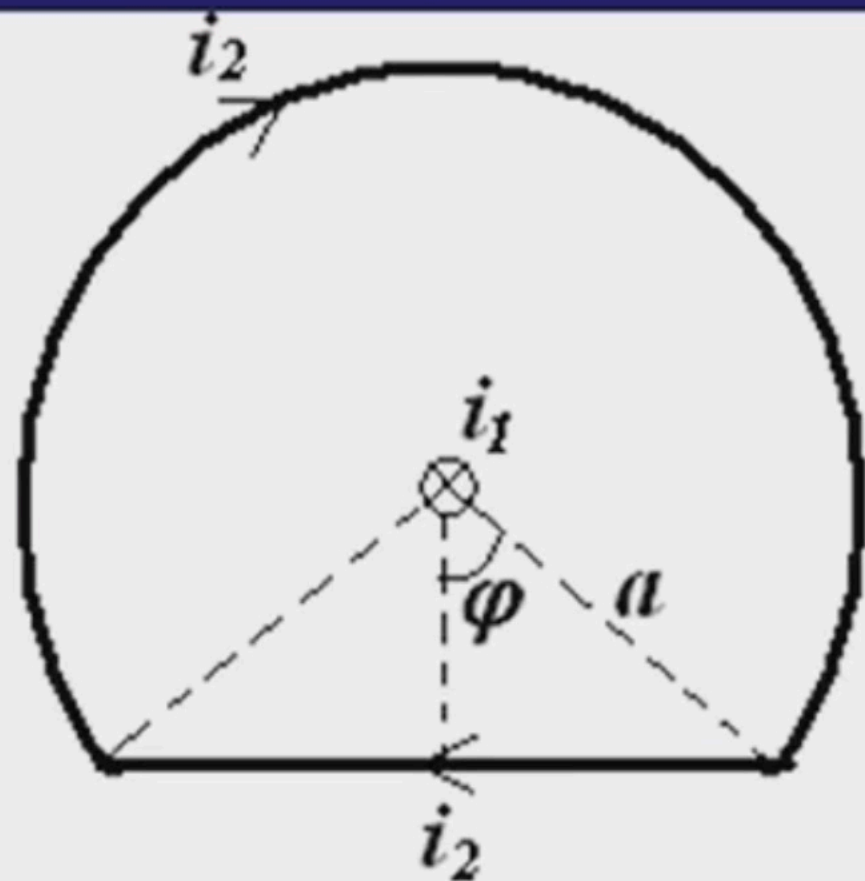


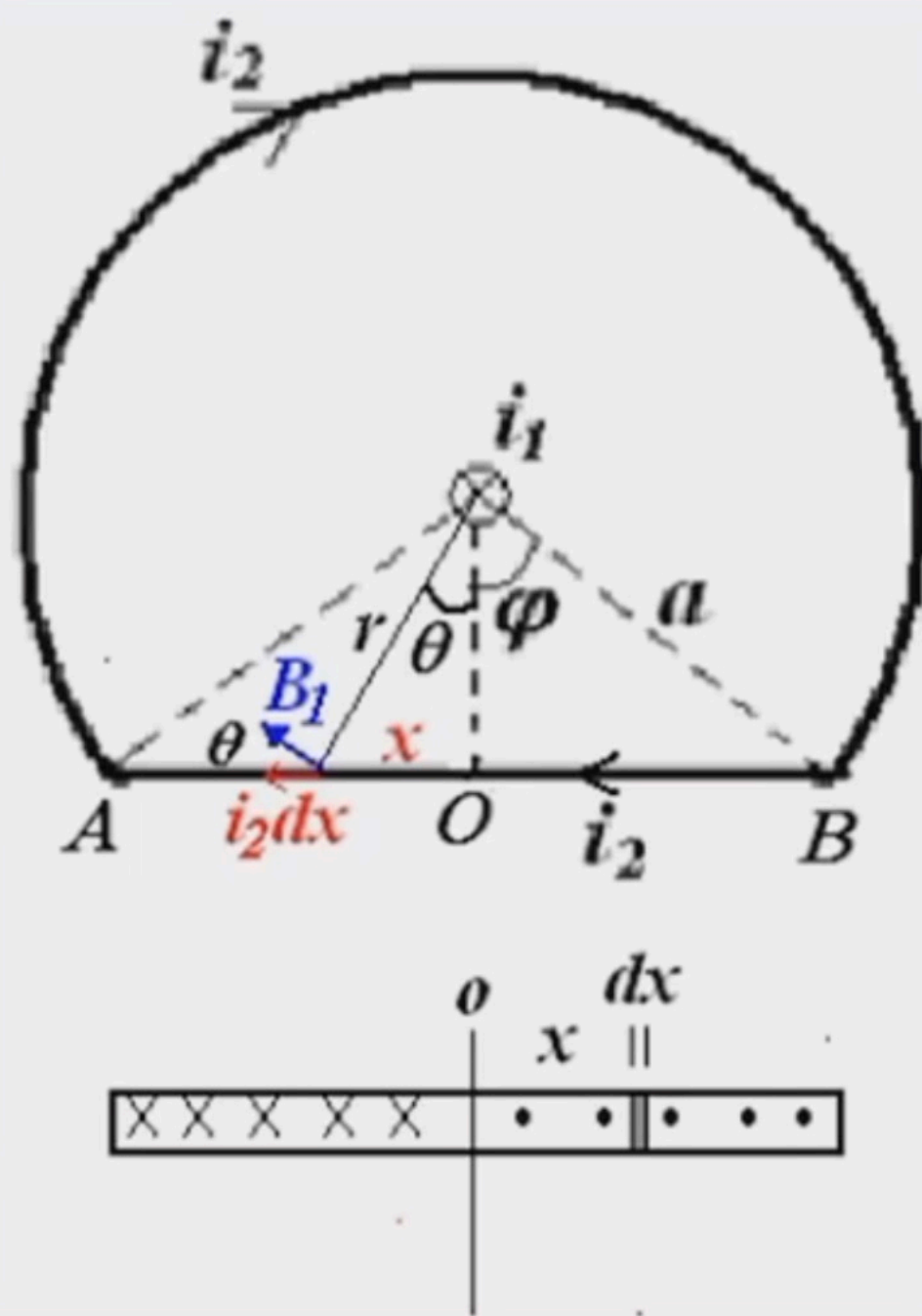
Figure 10

$$d\vec{F} = i d\vec{s} \times \vec{B}, \quad dF = i_2 dx \cdot B \sin \theta = \frac{\mu_0 i_1 i_2 dx}{2\pi r} \sin \theta$$

$$d\tau = x \cdot dF = \frac{\mu_0 i_1 i_2 \sin \theta}{2\pi r} x dx$$

$$x = a \cos \varphi \cdot \operatorname{tg} \theta, \quad dx = a \cos \varphi \cdot \frac{d\theta}{\cos^2 \theta}$$

$$r = \frac{a \cos \varphi}{\cos \theta}$$



$$d\tau = x \cdot dF = \frac{\mu_0 i_1 i_2 \sin \theta \cdot \cos \theta}{2\pi a \cos \varphi} a \cos \varphi \cdot \operatorname{tg} \theta \cdot \frac{a \cos \varphi}{\cos^2 \theta} d\theta$$

$$= \frac{\mu_0 i_1 i_2 a \cos \varphi}{2\pi} \operatorname{tg}^2 \theta d\theta$$

$$\tau = \int_{-\varphi}^{\varphi} \frac{\mu_0 i_1 i_2 a \cos \varphi}{2\pi} \operatorname{tg}^2 \theta d\theta$$

$$= \frac{\mu_0 i_1 i_2 a \cos \varphi}{2\pi} \int_{-\varphi}^{\varphi} \left[\frac{1}{\cos^2 \theta} - 1 \right] d\theta$$

$$= \frac{\mu_0 i_1 i_2 a}{\pi} (\sin \varphi - \varphi \cos \varphi)$$

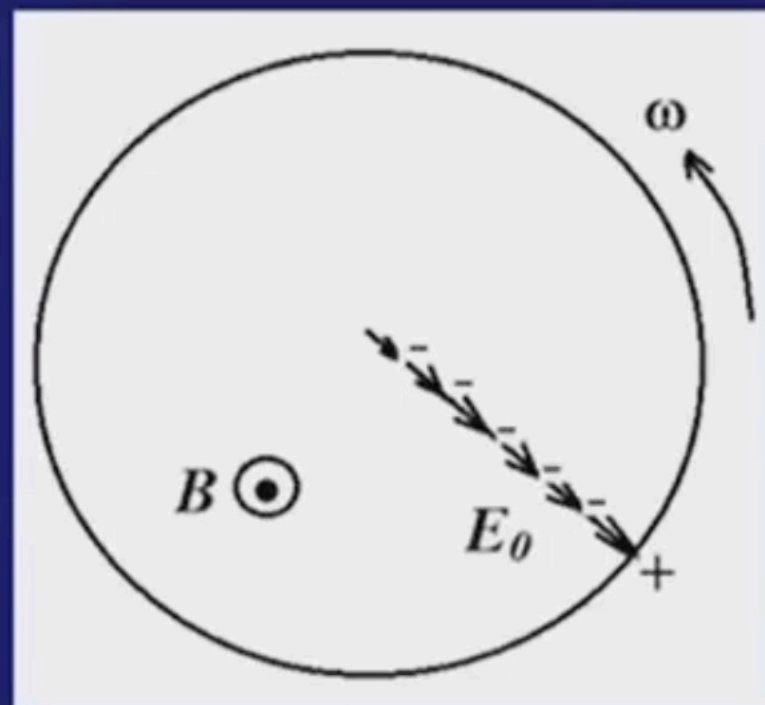
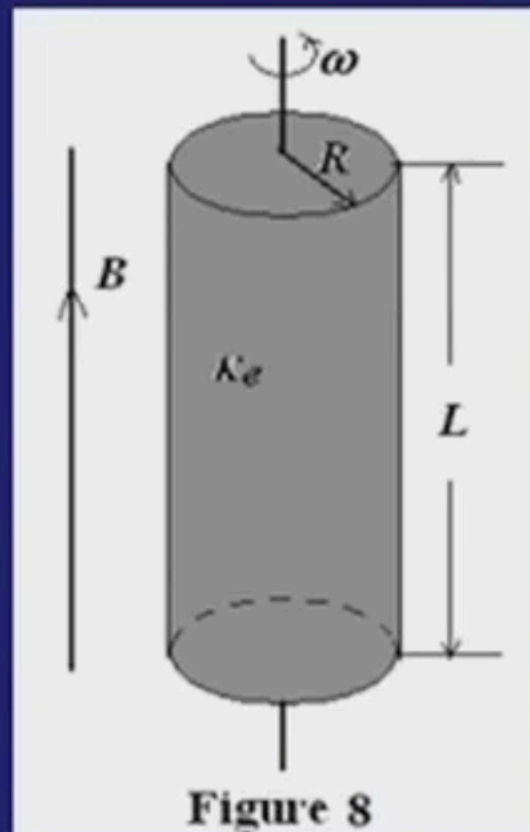
2. (18%) As shown in Fig. 8, a non-magnetic insulating cylinder of radius R , length L , is made of uniform, isotropic dielectric (均匀的、各向同性的电介质) with a dielectric constant κ_e . At beginning, the cylinder does not carry any charges. When the cylinder rotates around its axis with an angular speed ω , and is placed in a magnetic field B , which is applied along the axis of cylinder, please calculate.

(1). The magnitude and direction of polarization, P , in the cylinder.

(2). The volume induced-charge density, ρ_e' , in the cylinder.

(3). The induced-surface-charge density, σ_e' at $r=R$.

(4). The magnetic moment of this cylinder.



$$\vec{E}_0 = \vec{v} \times \vec{B}, \quad E_0 = r\omega B, \quad E = \frac{E_0}{\kappa_e} = \frac{r\omega B}{\kappa_e}$$

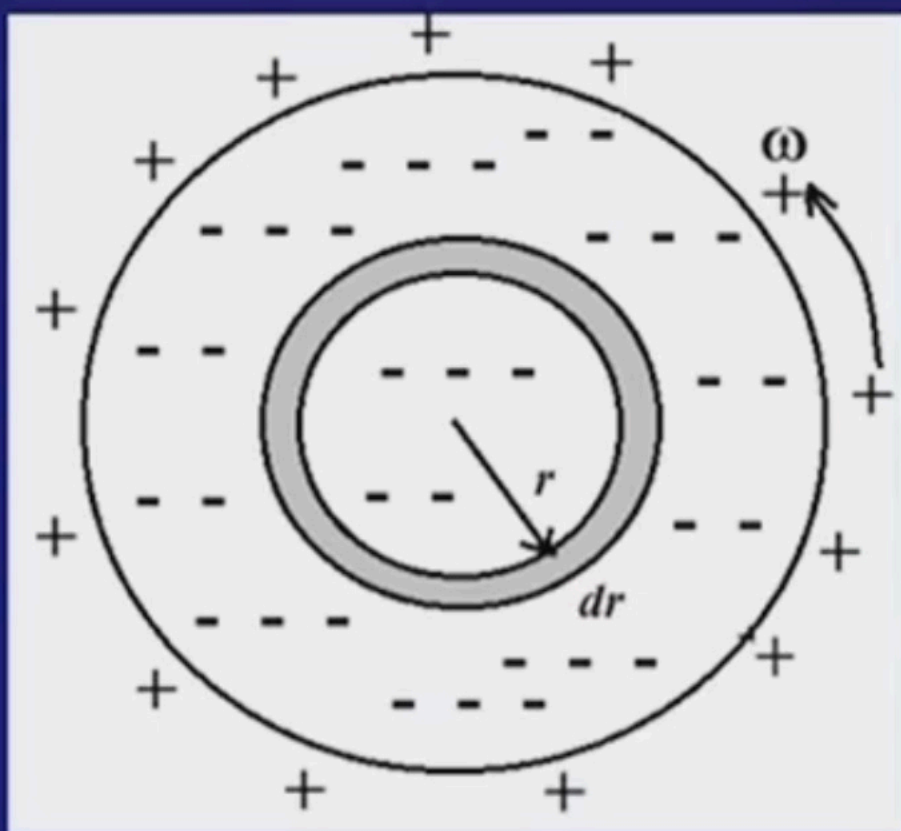
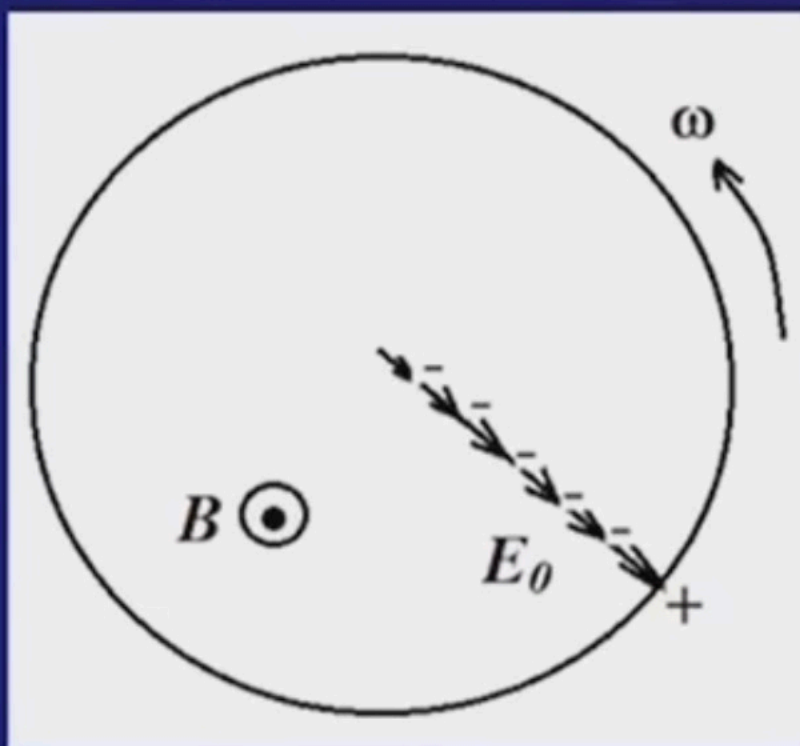
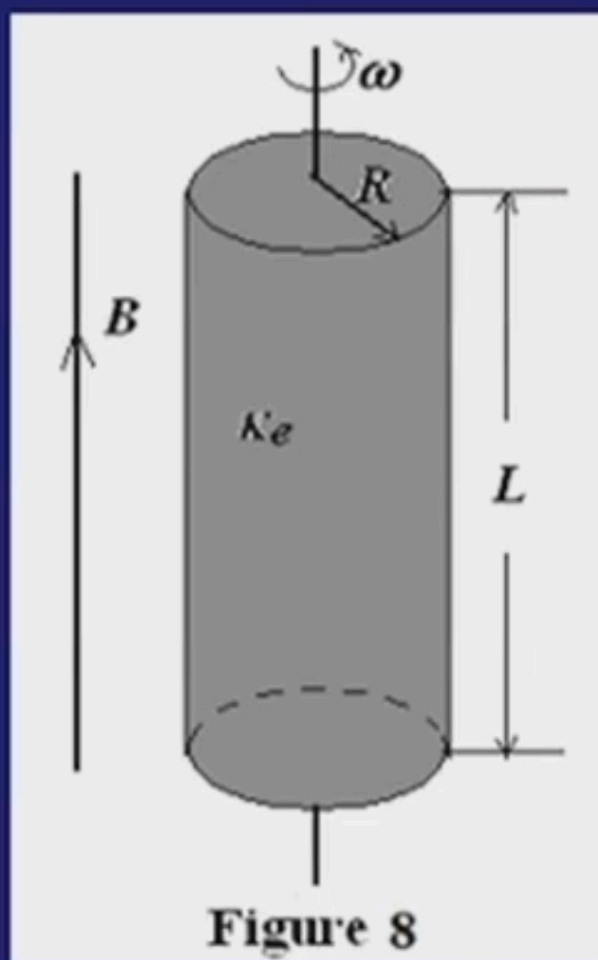
$$(1). \vec{P} = \chi_e \epsilon_0 \vec{E} = (\kappa_e - 1) \epsilon_0 \vec{E},$$

$$P = (\kappa_e - 1) \frac{\epsilon_0}{\kappa_e} r\omega B = \frac{\kappa_e - 1}{\kappa_e} \epsilon_0 r\omega B$$

$$(2). \rho_e' = -\nabla \cdot \vec{P}$$

$$\rho_e'(r) = -\frac{1}{r} \frac{\partial}{\partial r} (rP_r) = -\frac{2(\kappa_e - 1)}{\kappa_e} \epsilon_0 \omega B$$

$$(3). r = R, \quad \sigma_e' = \vec{P} \cdot \vec{n} = \frac{\kappa_e - 1}{\kappa_e} \epsilon_0 R\omega B$$



$$(2). \rho_e'(r) = -\frac{1}{r} \frac{\partial}{\partial r} (rP_r) = -\frac{2(\kappa_e - 1)}{\kappa_e} \varepsilon_0 \omega B$$

$$(3). \sigma_e' = \vec{P} \cdot \vec{n} = \frac{\kappa_e - 1}{\kappa_e} \varepsilon_0 R \omega B$$

$$(4). di_v = \frac{dq}{T} = \frac{\rho_e' 2\pi r L dr}{2\pi / \omega}$$

$$= \frac{2(\kappa_e - 1)}{\kappa_e} \varepsilon_0 L \omega^2 B r dr$$

$$\mu_v = \int_0^R \pi r^2 di_v = \int_0^R \pi r^2 \frac{2(\kappa_e - 1)}{\kappa_e} \varepsilon_0 L \omega^2 B r dr$$

$$= \frac{\kappa_e - 1}{2\kappa_e} \pi \varepsilon_0 L \omega^2 B R^4$$

$$\mu_s = \pi R^2 i_s = \pi R^2 \frac{\sigma_e(R) \cdot 2\pi R L}{2\pi / \omega}$$

$$= \frac{\kappa_e - 1}{\kappa_e} \pi \varepsilon_0 L \omega^2 B R^4$$

$$\mu_{total} = \mu_s - \mu_v = \frac{\kappa_e - 1}{2\kappa_e} \pi \varepsilon_0 L \omega^2 B R^4$$

3. (20%) As shown in Fig. 9(a), a thin circular disk of radius a , thickness d , and conductivity (电导率) σ is placed in a uniform time varying magnetic field $B(t)$.

- (1) Neglecting the magnetic field of the eddy currents, what is the current induced in a thin circular filament at radius r of thickness dr .
- (2) What power is dissipated (损耗) in this thin current loop?
- (3) How much power is dissipated in the whole disk?
- (4) If the disk is cut up into N smaller circular disks with negligible wastage (缝隙), as shown in Fig. 9(b), what is the approximate radius of each smaller disk?
- (5) If these N smaller disks are laminated (叠压) together to form a thin disk of closely packed cylindrical wires, what is the power dissipated?

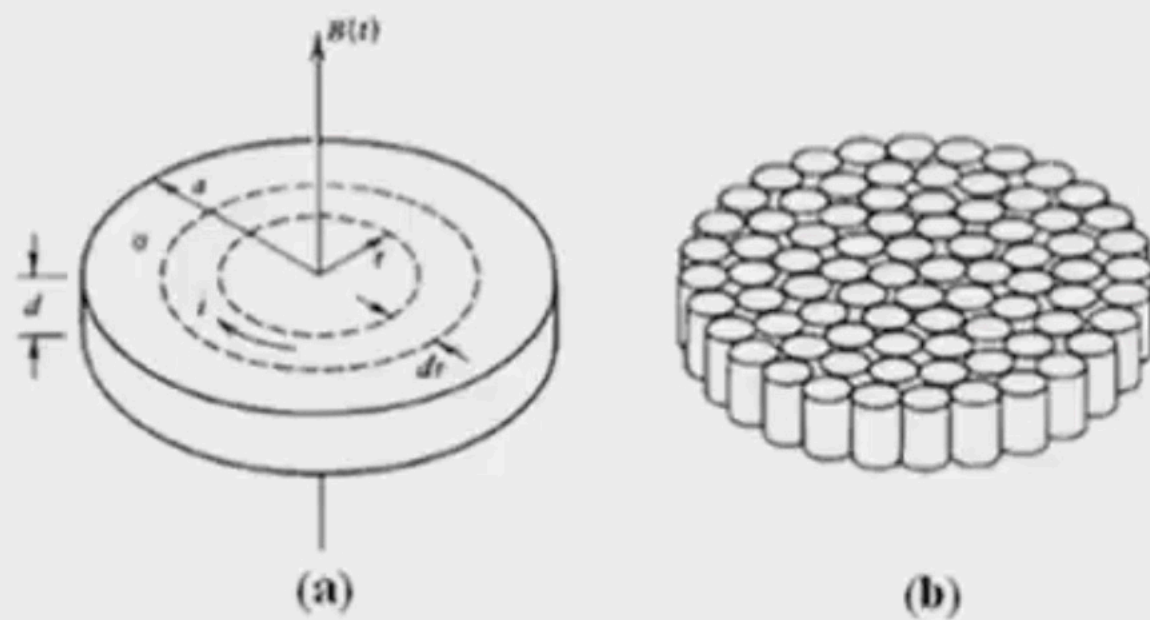


Figure 9

$$(1). \oint \vec{E} \cdot d\vec{l} = - \iint \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A}$$

$$E \cdot 2\pi r = - \frac{dB}{dt} \cdot \pi r^2, \quad E = - \frac{1}{2} r \frac{dB}{dt}$$

$$j = \sigma E = \frac{1}{2} \sigma r \frac{dB}{dt}$$

$$di = j \cdot d \cdot dr = \frac{1}{2} r \sigma d \frac{dB}{dt} dr$$

$$(2). \quad dP = \varepsilon di = E \cdot 2\pi r di = \frac{1}{2} r \frac{dB}{dt} \cdot 2\pi r \cdot \frac{1}{2} r \sigma d \frac{dB}{dt} dr$$

$$= \frac{1}{2} \pi \sigma d \left(\frac{dB}{dt} \right)^2 r^3 dr$$

3. (20%) As shown in Fig. 9(a), a thin circular disk of radius a , thickness d , and conductivity (电导率) σ is placed in a uniform time varying magnetic field $B(t)$.

- (1) Neglecting the magnetic field of the eddy currents, what is the current induced in a thin circular filament at radius r of thickness dr .
- (2) What power is dissipated (损耗) in this thin current loop?
- (3) How much power is dissipated in the whole disk?
- (4) If the disk is cut up into N smaller circular disks with negligible wastage (缝隙), as shown in Fig. 9(b), what is the approximate radius of each smaller disk?
- (5) If these N smaller disks are laminated (叠压) together to form a thin disk of closely packed cylindrical wires, what is the power dissipated?

$$(3). \quad P = \int_0^a \frac{1}{2} \pi \sigma d \left(\frac{dB}{dt} \right)^2 r^3 dr = \frac{1}{8} \pi \sigma d \left(\frac{dB}{dt} \right)^2 a^4$$

$$(4). \quad \pi a^2 = N(\pi R^2) \quad R = \sqrt{\frac{1}{N}} a$$

$$(5). \quad P_{small} = \frac{1}{8} \pi \sigma d \left(\frac{dB}{dt} \right)^2 \cdot \frac{1}{N^2} a^4$$

$$P_{total} = N \cdot P_{small} = \frac{1}{N} P$$

